SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE
(AUTONOMOUS)
(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)
Accredited by NAAC with "A+" Grade.
Recognised as Scientific and Industrial Research Organisation
SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Estd. 1980

Regulation:R23	IV/IV-B.Tech. I -Semester

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

COURSE STRUCTURE
(With effect from 2023-24 admitted Batch onwards)

Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23AD4101	Generative AI Techniques	PC	3	0	0	3	30	70	100
B23HS4101	Human Resource & Project Management	HS	2	0	0	2	30	70	100
#PE-IV	Professional Elective-IV	PE	3	0	0	3	30	70	100
#PE-V	Professional Elective-V	PE	3	0	0	3	30	70	100
#OE-III	Open Elective-III	OE	3	0	0	3	30	70	100
#OE-IV	Open Elective-IV	OE	3	0	0	3	30	70	100
B23AD4108	Prompt Engineering OR Any of the APSCHE offered skill-oriented course	SEC	0	1	2	2	30	70	100
B23MC4101	Constitution of India	MC	2	0	0	-	30	-	30
B23AD4109	Evaluation of Industry Internship /Mini Project	PR	-	-	-	2	-	50	50
		TOTAL	19	1	2	21	240	540	780

	Course Code	Course
#PE-IV	B23CS4102	Software Architecture & Design Patterns
	B23IT4104	Blockchain Technology & Applications
	B23AD4102	DevOps
	B23AD4103	Architecture for Management of Large Datasets
	B23AD4104	Any of the12-Week SWAYAM/NPTEL Course recommended by the BoS
#PE-V	B23CS4107	Agile Methodologies
	B23AD4105	Robotic Process Automation
	B23AD4106	Reinforcement Learning
	B23AM4106	High Performance Computing
	B23AD4107	Any of the12-Week SWAYAM/NPTEL Course recommended by the BoS
#OE-III	Student has to study one Open Elective offered by CE or ECE or EEE or ME or S&H from the list enclosed.	
#OE-IV	Student has to study one Open Elective offered by CE or ECE or EEE or ME or S&H from the list enclosed.	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AD4101	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

Generative AI Techniques

for AIDS

Course Objectives: This course is designed to:

1	Introduce the basics of Generative AI, Text Generation, the process of generating videos and GAN & variants.
2	Understand Prompt Engineering and different open-source models.

Course Outcomes: Upon the completion of the course students will be able to:

S. No	Outcome	Knowledge Level
1	**Explain** the basic concepts of generative models in AI and use cases	**K2**
2	**Analyze** various Generative Models for Text generation.	**K3**
3	**Use** different Generative Models For images transformation.	**K3**
4	**Analyze** GPT along with prompt engineering	**K3**
5	**Apply** various Open-Source Models and Programming Frameworks	**K3**

SYLLABUS

UNIT-I (10Hrs)	**Introduction To Gen AI:** Historical Overview of Generative modeling, Difference between Gen AI and Discriminative Modeling, Importance of generative models in AI and Machine Learning, Types of Generative models, GANs, VAEs, autoregressive models and Vector quantized Diffusion models, Understanding if probabilistic modeling and generative process, Challenges of Generative Modeling, Future of Gen AI, Ethical Aspects of AI, Responsible AI, Use Cases.
UNIT-II (8 Hrs)	**Generative Models for Text:** Introduction, Natural Language Processing, Sequence to sequence models: RRN, LSTM, **Transformer architecture:** Types of architecture, Transformer components: Tokenization, Positional Embeddings Encoder layer, Residual connections, Decoder, Attention mechanisms, NLP: Models like BERT and GPT models.
UNIT-III (10 Hrs)	**Generation of Images:** Introduction to computer vision, Representation learning: Auto encoders and variational autoencoders. **Transforming Images with GenAI**: Generative modelling, Generative adversarial networks: GAN variants, Diffusion model, Flow model, Evaluation metrics for image generative model.
UNIT-IV (10 Hrs)	**ChatGPT:** Introduction to OpenAI: OpenAI image models, rise OpenAI Whisper, OpenAI playground, OpenAI Cookbook, OpenAI fine tuning ChatGPT plugins, ChatGPT Enterprise, pricing, Limitations of ChatGPT, ChatGPT future **Prompt engineering:** Importance of prompt, prompt engineering process, setting objectives, prompt design, Evaluation of responses, prompt refining, prompt

	templating.
UNIT-V (8 Hrs)	**Open-Source Models and Programming Frameworks:** Rise of Open-source Generative Models: closed source models Vs Open-source models, Hugging Face, Open-source Gen AI models and their Impact, Comparison of Open source LLM Models, Fine-Tuned Models Vs Pre-Trained Models **Programming LLMs:** OpenAI with Python, Exploring LLM frameworks, Introduction to LangChain, Components of LangChain, Applications of LangChain, Agents, Retrieval Augmented Generation.

Textbooks:

1.	Karthikeyan Sabesan,” Generative AI for Everyone “BPB publications
2.	Altaf Rehmani, “Generative AI for Everyone”, BlueRose One, 2024.

Reference Books:

1.	David Foster,” Generative Deep Learning”, O’Reily Books, 2024.
2.	Denis Rothman, “Transformers for Natural Language Processing and Computer Vision”, Third Edition, Packt Books, 2024.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS4101	**HS**	**2**	**--**	**--**	**2**	**30**	**70**	**3 Hrs.**

HUMAN RESOURCE & PROJECT MANAGEMENT

(Common to CSE, AIML, IT, AIDS, CSG, CSIT)

Course Objectives:

1.	Explain the nature, scope, functions of HRM, and processes of HR planning, recruitment, and selection.
2.	Explain HRD concepts, training methods, performance appraisal systems, and career development practices.
3.	Explain the basic concepts of project management, including project life cycle, resource management, and monitoring techniques.
4.	Explain project selection and appraisal methods using financial and strategic evaluation tools.
5.	Explain project implementation processes, organizational structures, and performance evaluation methods.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Explain HRM functions, HR planning, and recruitment–selection procedures.	K2
2.	Explain HRD practices, training techniques, and performance appraisal methods.	K2
3.	Explain project management concepts, project networks, and monitoring processes.	K2
4.	Explain project appraisal techniques such as SWOT analysis, payback period, and NPV.	K2
5.	Explain project implementation strategies, organizational forms, and performance evaluation processes.	K2

SYLLABUS

UNIT-I (10Hrs)	**HRM:** Nature, Scope, Concept of HRM - Functions of HRM - Role of HR manager – Emerging trends in HRM – E-HRM - HR audit models – ethical aspects of HRM. HR Planning – Job Design- Recruitment - Sources of recruitment -Selection- Selection Procedure.
UNIT-II (10 Hrs)	**HRD**: HR accounting – Models - Concept of Training and Development –- Methods of Training. Performance Appraisal: Importance – Methods of performance appraisal – Career Development and Counseling – group interaction.
UNIT-III (10 Hrs)	**Basics of Project Management:** Concept–resource management- Project environment – Types of Projects –project networks-DPR- Project life cycle – Project proposals –

	Monitoring project progress – Project appraisal and Project selection-80-20 rules.
UNIT-IV (10 Hrs)	**Project Selection and Appraisal:** Brainstorming and concept evolution, Project selection and evaluation, Selection criteria and models, Types of appraisals, SWOT analysis, Cash flow analysis, Payback period, and Net present value.
UNIT-V (10 Hrs)	**Project Implementation and Review:** Identify various project types and apply appropriate management strategies for each. Forms of project organization – project planning – project control – human aspects of project management – prerequisites for successful project implementation – project review.
Textbooks:	
1.	Sharon Pande and Swapnalekha Basak, Human Resource Management, Text and Cases, Vikas Publishing, 2e, 2016.
2.	K. Nagarajan, Project Management, New Age International Publishers, 8e, 2017
Reference Books:	
1.	Subba Rao P: ―Personnel and Human Resource Management-Text and Cases‖, Himalaya Publications, Mumbai, 2013.
2.	Prasanna Chandra, ―Projects, Planning, Analysis, Selection, Financing Implementation and Review‖, Tata McGraw Hill Company Pvt. Ltd., New Delhi 1998
3.	Vasanth Desai, Project Management, 4th edition, Himalaya Publications 2018.
4.	Lalithabalakrishnan and Gowri, Project Management, Himalaya publishing house, New Delhi 2022
5.	K Aswathappa: ―Human Resource and Personnel Management‖, Tata McGraw Hill, New Delhi, 2013
e-Resources	
1.	Robert L. Mathis, John H. Jackson, Manas Ranjan Tripathy, Human Resource Management, Cengage Learning 2016.
2.	Stewart R. Clegg, Torgeir Skyttermoen, Anne Live Vaagaasar, Project Management, Sage Publications, 1e, 2021.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4102	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

SOFTWARE ARCHITECTURE & DESIGN PATTERNS		
(Common to CSE, AIDS, CSBS, CSD)		
Pre-requisites: OOP,SE,OOAD		
Course Objectives: Students are expected		
1.	Understand the basic concepts to identify state behaviour of real world objects	
2.	Apply Object Oriented Analysis and Design concepts to solve complex problems	
3.	Construct various UML models using the appropriate notation for specific problem context	
4.	Design models to Show the importance of systems analysis and design in solving complex problems using case studies	
5.	Study Pattern Oriented approach for real world problems	
Course OutComes: At the end of the course students will be able to		
S. No	**outcome**	KL
1.	**Explain** the concepts of design patterns and object-oriented development.	2
2.	**Use** object-oriented analysis techniques to gather requirements and identify conceptual classes and relationships in a system.	3
3.	**Demonstrate** structural design patterns such as Adapter, Bridge, Composite, Decorator, Facade, Flyweight, and Proxy.	3
4.	**Illustrate** the MVC architectural pattern in designing interactive systems.	3
5.	**Apply** distributed object system concepts including client–server architecture, Java RMI, and web services.	3

SYLLABUS

UNIT-I (10 Hrs)	Introduction: design pattern, Describing design patterns, the catalog of design pattern, organizing the catalog, how design patterns solve design problems, how to select a design pattern, how to use a design pattern. What is object oriented development-key concepts of object oriented design other related concepts, benefits and drawbacks of the paradigm
UNIT-II (10 Hrs)	Analysis a System: Overview of the analysis phase, stage 1 gathering the requirements functional requirements specification, defining conceptual classes and relationships, using the knowledge of the domain Design and Implementation, discussions and further reading
UNIT-III (10 Hrs)	Design Pattern Catalog: Structural patterns, Adapter, bridge, composite, decorator, facade, Flyweight, proxy.
UNIT-IV (10 Hrs)	Interactive systems and the MVC architecture: Introduction The MVC architectural pattern, analyzing a simple drawing program designing the system, designing of the

	subsystems, getting into implementation, implementing undo operation drawing incomplete items, adding a new feature pattern based solutions
UNIT-V (10 Hrs)	Designing with Distributed Objects: Client server system, java remote method invocation, implementing an object oriented system on the web, Web services (SOAP, Restful), Enterprise Service Bus

Text books:	
1.	Object Oriented Analysis, Design and Implementation, Brahma Dathan, Sarnath Rammath , Universities Press, 2013
2.	Design Patterns, Erich Gamma, Richard Helan, Ralph Johman, John Vlissides, PEARSON Publication, 2013
Reference books:	
1.	Frank Bachmann, Regine Meunier, Hans Rohnert “Pattern Oriented Software Architecture”, Volume 1, 1996.
2.	William J Brown et al., "Anti Patterns: Refactoring Software, Architectures and Projects in Crisis", John Wiley, 1998
E-Resources:	
1.	Object Oriented Analysis and Design By Prof. Partha Pratim Das, Prof. Samiran Chattopadhyay, Prof. Kausik Datta | IIT Kharagpur https://onlinecourses.nptel.ac.in/noc21_cs57/preview
2.	Object Oriented System Development Using UML, Java And Patterns, IIT Kharagpur Prof. Rajib Mall https://nptel.ac.in/courses/106105224

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4104	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

BLOCKCHAIN TECHNOLOGY & APPLICATIONS

(Common to IT,AIDS, CSBS)

Course Objectives: This course is designed to:

1.	Understand the fundamental concepts of blockchain technology, its architecture, and working principles.
2.	Explain different types of blockchain systems and consensus mechanisms used in distributed environments.
3.	Analyze public, private, and consortium blockchain platforms and their real-world implementations.
4.	Explain and analyze cryptocurrency systems and smart contract mechanisms.
5.	Analyze security, privacy challenges, and performance issues in blockchain technology.

Course Outcomes: Upon the completion of the course students will be able to:

S.No	Outcome	Knowledge Level
1.	**Explain** blockchain fundamentals, architecture, and cryptographic concepts	K2
2.	**Explain** smart contracts and cryptocurrency transaction mechanisms in blockchain platforms	K3
3.	**Analyze** private and consortium blockchain systems and permissioned blockchain algorithms	K2
4.	**Analyze** security, privacy challenges and applications of blockchain in emerging real-world scenarios.	K3
5.	**Analyze** blockchain case studies and implement blockchain applications using python and hyperledger fabric.	K3

SYLLABUS

UNIT-I	**Fundamentals of Blockchain:** Introduction, Origin of Blockchain, Blockchain Solutions, Components of Blockchain, Block in a Blockchain, The Technology and the Future. Blockchain Types and Consensus Mechanism: Introduction, Decentralization and Distribution, Types of Blockchain, Consensus Protocol. Cryptocurrency: Bitcoin, Altcoin and Token: Introduction, Bitcoin and the Cryptocurrency, Cryptocurrency Basics, Types of Cryptocurrencies, Cryptocurrency Usage.
UNIT-II	**Public Blockchain System:** Introduction, Public Blockchain, Popular Public Blockchains, The Bitcoin Blockchain, Ethereum Blockchain. Smart Contracts: Introduction, Smart Contract, Characteristics of a Smart Contract, Types of Smart Contracts, Types of Oracles, Smart Contracts in Ethereum, Smart Contracts in Industry.

UNIT-III	**Private Blockchain System:** Introduction, Key Characteristics of Private Blockchain, Private Blockchain, Private Blockchain Examples, Private Blockchain and Open Source, E-commerce Site Example, Various Commands (Instructions) in E-commerce Blockchain, Smart Contract in Private Environment, State Machine, Different Algorithms of Permissioned Blockchain, Byzantine Fault, Multichain. Consortium Blockchain: Introduction, Key Characteristics of Consortium Blockchain, Need of Consortium Blockchain, Hyperledger Platform, Overview of Ripple, Overview of Corda.
UNIT-IV	**Security in Blockchain**: Introduction, Security Aspects in Bitcoin, Security and Privacy Challenges of Blockchain in General, Performance and Scalability, Identity Management and Authentication, Regulatory Compliance and Assurance, Safeguarding Blockchain Smart Contract (DApp), Security Aspects in Hyperledger Fabric. **Applications of Blockchain:** Introduction, Blockchain in Banking and Finance, Blockchain in Education, Blockchain in Energy, Blockchain in Healthcare, Blockchain in Real-estate, Blockchain in Supply Chain, The Blockchain and IoT. Limitations and Challenges of Blockchain.
UNIT-V	**Blockchain Case Studies:** Case Study 1 – Retail, Case Study 2 – Banking and Financial Services, Case Study 3 – Healthcare, Case Study 4 – Energy and Utilities. Blockchain Platform using Python: Introduction, Learn How to Use Python Online Editor, Basic Programming Using Python, Python Packages for Blockchain. Blockchain platform using Hyperledger Fabric: Introduction, Components of Hyperledger Fabric Network, Chain codes from Developer.ibm.com, Blockchain Application Using Fabric Java SDK.
Textbooks:	
1.	"Blockchain Technology", Chandramouli Subramanian, Asha A.George, Abhilasj K A, Meena Karthikeyan , Universities Press.
2.	Blockchain for Business, Jai Singh Arun, Jerry Cuomo, Nitin Gauar, Pearson Addition Wesley
Reference Books:	
1.	Blockchain Blue print for Economy, Melanie Swan, SPD Oreilly.
e-Resources:	
1	NPTEL: Blockchain Architecture Design and Use Cases – https://nptel.ac.in
2	Ethereum Documentation – https://ethereum.org
3	Hyperledger Documentation – https://hyperledger.org

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AD4102	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

DevOps

(Common to AIDS & CSIT)

Course Objectives: This course is designed to:

1.	To describe the agile relationship between development and IT operations
2.	To understand the skill sets and high-functioning teams involved in DevOps and related methods to reach a continuous delivery capability
3.	To implement automated system update and DevOps lifecycle

Course Outcomes: At the end of the course students will be able to

S. No	Outcome	Knowledge Level
1.	**Describe** the DevOps lifecycle, Agile principles, and the architecture of CI/CD pipelines.	K2
2.	**Execute** source code management tasks using Git and perform code quality analysis with SonarQube	K3
3.	**Construct** automated build pipelines and manage Jenkins architecture for Continuous Integration.	K3
4.	**Deploy** containerized applications using Docker and automate testing using Selenium	K3
5.	**Orchestrate** application deployments using Ansible and Kubernetes resource management.	K3

SYLLABUS

UNIT-I (10Hrs)	**Introduction to DevOps:** Introduction to SDLC, Agile Model. Introduction to Devops. DevOps Features, DevOps Architecture, DevOps Lifecycle, Understanding Workflow and principles, Introduction to DevOps tools, Build Automation, Delivery Automation, Understanding Code Quality, Automation of CI/ CD. Release management, Scrum, Kanban, delivery pipeline, bottlenecks, examples
UNIT-II (8 Hrs)	**Source Code Management (GIT):** The need for source code control, The history of source code management, Roles and code, source code management system and migrations. What is Version Control and GIT, GIT Installation, GIT features, GIT workflow, working with remote repository, GIT commands, GIT branching, GIT staging and collaboration. **Unit Testing - Code Coverage**: Junit, nUnit & Code Coverage with Sonar Qube, SonarQube - Code Quality Analysis.
UNIT-III (10 Hrs)	**Build Automation - Continuous Integration (CI):** Build Automation, What is CI Why CI is Required, CI tools, Introduction to Jenkins (With Architecture), jenkins workflow, **Jenkins master slave architecture**, Jenkins Pipelines, PIPELINE BASICS - Jenkins Master,

	Node, Agent, and Executor Freestyle Projects & Pipelines, Jenkins for Continuous Integration, Create and Manage Builds, User Management in Jenkins, Schedule Builds, Launch Builds on Slave Nodes.
UNIT-IV (10 Hrs)	**Continuous Delivery (CD):** Importance of Continuous Delivery, Continuous Deployment CD Flow, Containerization with Docker: Introduction to Docker, Docker installation, Docker commands, Images & Containers, DockerFile, running containers, Working with containers and publish to Docker Hub.
	Testing Tools: Introduction to Selenium and its features, JavaScript testing
UNIT-V (8 Hrs)	**Configuration Management - ANSIBLE:** Introduction to Ansible, Ansible tasks, Roles, Jinja templating, Vaults, Deployments using Ansible. Containerization Using Kubernetes (Openshift): Introduction to Kubernetes Namespace & Resources, CI/CD - On OCP, BC, DC & ConfigMaps, Deploying Apps on Openshift Container Pods. Introduction to Puppet master and Chef.

Textbooks:	
1.	Agarwal, Gaurav., Modern DevOps Practices: Implement, secure, and manage applications on the public cloud by leveraging cutting-edge tools, 2nd Edition, Packt Publishing, 2024.
2.	Verma, Saurabh., Learning DevOps: Accelerate your organization's digitaltransformation by adopting DevOps practices, 2nd Edition, Packt Publishing, 2019.
Reference Books:	
1.	Len Bass, Ingo Weber, Liming Zhu. DevOps: A Software Architect's Perspective. Addison Wesley; ISBN-10
2.	Gene Kim Je Humble, Patrick Debois, John Willis. The DevOps Handbook, 1st Edition, IT Revolution Press, 2016.
3.	Verona, Joakim Practical DevOps, 1st Edition, Packt Publishing, 2016.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AD4103	PE	3	--	--	3	30	70	3 Hrs.

ARCHITECTURE FOR MANAGEMENT OF LARGE DATASETS

for AIDS

Course Objectives: This course is designed to

1.	Cover the fundamentals of Hadoop and administering the Hadoop
2.	Cover the fundamentals of Hive and Spark. Storm
3.	fundamentals of digital data and language models for graph

Course Outcomes: Upon the completion of the course students will be able to:

S. No	Outcome	Knowledge Level
1.	**Understand** the fundamentals of Hadoop technology	K2
2.	**Understand** the Administration of Hadoop	K2
3.	**Understand** the technology Hive	K2
4.	**Demonstrate** Spark and Understand of storm technology	K3
5.	**Demonstration** of building large language Models	K3

SYLLABUS

UNIT-I (10Hrs)	**Hadoop, HDFS and Setting up a Hadoop Cluster**: Why Hadoop, Why not RDBMS, RDMBS vs Hadoop, HDFS Daemons, Anatomy of File Read, Anatomy of File Write, Replica Placement Strategy, Working with HDFS commands, Setting up a Hadoop Cluster-Cluster Specification, Cluster Setup and Installation, Hadoop Configuration, Security
UNIT-II (8 Hrs)	**Terminologies and Administering Hadoop**: Terminologies used in Big Data Environments - In-memory analytics, In-database processing, Symmetric Multiprocessor System (SMP), Massively Parallel Processing, Difference between Parallel and Distributed Systems, Shared Nothing Architecture, Administering Hadoop-Administering HDFS, Monitoring and Maintenance
UNIT-III (10 Hrs)	**Hive and Architecture of Graph**: What is Hive, Hive Architecture, Hive Data Types, Hive File Format, HiveQL, Tables, Querying Data, User Defined Functions (UDF), Architecture of Graph
UNIT-IV (10 Hrs)	**Spark and Architecture of Storm**: Installing Spark, Spark Applications, Jobs, Stages, Tasks, Resilient Distributed Datasets, Shared Variables, Anatomy of a Spark Job Run, Executors and Cluster managers, Architecture of Storm
UNIT-V	**Classification of Digital Data and Large Language Models for Graphs**: Classification

(8 Hrs)	of Digital Data - Structured Data, Semi-structured Data and Unstructured Data, Understanding large language models-What is an LLM, Applications of LLMs, Stages of building and using LLMs, Introducing the transformer architecture, Utilizing large datasets, A closer look at the GPT architecture, Building a large language model.
Textbooks:	
1.	Seema Acharya and Suhashini Chellappan, “Big Data Analytics”, 2nd Ed, Wiley, 2015
2.	Tom White, “Hadoop: The Definitive Guide”, 4th Ed, O’reilly, 2015
3	V. K. Jain, Big Data and Hadoop, Khanna Book Publishing, 2020
4	Build a Large Language Model (From Scratch), Sebastian Raschka, September 2024 ISBN 9781633437166
Reference Books:	
1.	Michael Minelli, Michelle Chambers and Ambiga Dhiraj, “Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today’s Businesses”, Wiley, 2013
2.	P. J. Sadalage, M. Fowler, “NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence”, Addison-Wesley Professional, 2012
3.	Storm https://cds.iisc.ac.in/wp-content/uploads/DS256.2017.Storm_.Tutorial.pdf
4	Giraph https://cds.iisc.ac.in/wp-content/uploads/DS256.2017.L12.Pregel.pdf

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4107	PE	3	--	--	3	30	70	3 Hrs.

AGILE METHODOLOGIES

(Common to CSE, AIML, AIDS, CSBS, IT)

Pre-requisites: Software Engineering

Course Objectives: Students are expected

1.	Gain deep comprehension of Agile philosophy, principles, and values to support high-quality software delivery.
2.	Employ different Agile frameworks (Scrum, XP, Lean, Kanban) to real-world projects for improved adaptability, productivity, and customer satisfaction.

Course Out Comes: At the end of the course students will be able to

S. No	OUTCOME	KL
1.	Apply agile values and principles to foster team collaboration and adaptability in projects.	K3
2.	Demonstrate the 12 agile principles to direct project delivery in changing contexts.	K3
3.	Use Scrum roles, events, and artifacts to oversee iterative development cycles.	K3
4.	Model XP practices to boost software quality and team responsiveness.	K3
5.	Determine Lean and Kanban principles to detect waste and streamline development workflows.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Learning Agile:** Agile, Getting Agile into your brain, **Understanding Agile values**: No Silver Bullet, Agile to the Rescue. A fractured perspective, The Agile Manifesto, Understanding the Elephant, Where to Start with a New Methodology.
UNIT-II (10 Hrs)	**The Agile Principles:** The 12 Principles of Agile Software, The Customer Is Always Right, Delivering the Project, Better Project Delivery for the Ebook Reader Project. Communicating and Working Together, Project Execution—Moving the Project Along, Constantly Improving the Project and the Team. The Agile Project: Bringing All the Principles Together
UNIT-III (10 Hrs)	**SCRUM and Self-Organizing Teams:** The Rules of Scrum, Act I: I Can Haz Scrum, Everyone on a Scrum Team owns the Project, Status Updates Are for Social Networks!, The Whole Team Uses the Daily Scrum, Feedback and the Visibility-Inspection-Adaptation Cycle, The Last Responsible Moment, Sprinting into a Wall, Sprints, Planning, and Retrospectives. **Scrum Planning And Collective Commitment:** Not Quite Expecting the Unexpected,

	User Stories, Velocity, and Generally Accepted Scrum Practices, Victory Lap, Scrum Values Revisited.
UNIT-IV (10 Hrs)	**XP And Embracing Change:** Going into Overtime, The Primary Practices of XP, The Game Plan Changed, but We're Still Losing, The XP Values Help the Team Change Their Mindset, An Effective Mindset Starts with the XP Values, The Momentum Shifts, Understanding the XP Principles Helps You Embrace Change. **XP, Simplicity, and Incremental Design:** Code and Design, Make Code and Design Decisions at the Last Responsible Moment, Final Score.
UNIT-V (10 Hrs)	**Lean, Eliminating Waste, and Seeing the whole:** Lean Thinking, Creating Heroes and Magical Thinking. Eliminate Waste, Gain a Deeper Understanding of the Product, Deliver As Fast As Possible. **Kanban, Flow, and Constantly Improving:** The Principles of Kanban, Improving Your Process with Kanban, Measure and Manage Flow, Emergent Behavior with Kanban. **The Agile Coach:** Coaches Understand Why People Don't Always Want to Change. The Principles of Coaching.
Textbook:	
1.	**Andrew Stellman, Jennifer Greene, Learning Agile, O'Reilly, 2015.**
Reference books:	
1.	Andrew Stellman, Jennifer Green, Head first Agile, O'Reilly, 2017.
2.	Rubin K , Essential Scrum : A Practical Guide To The Most Popular Agile Process, Addison-Wesley, 2013
E-Resources:	
1.	https://www.agilealliance.org
2.	https://www.scrum.org
3.	https://www.scrumtrainingseries.com
4.	https://www.agilemanifesto.org
5.	https://www.scrumguides.org

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AD4105	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

ROBOTIC PROCESS AUTOMATION

for AIDS

Course Objectives: This course is designed to:

1.	Introduce RPA concepts, tools, and workflow development techniques.
2.	Develop skills in data handling, UI automation, and control flow.
3.	Enable bot deployment, monitoring, and exception handling.

Course Outcomes: Upon the completion of the course students will be able to:

S. No	Outcome	Knowledge Level
1.	**Summarize** the scope, components, and benefits of Robotic Process Automation.	K2
2.	**Apply** recording, sequences, and control flow to build simple automation workflows.	K3
3.	**Apply** UI controls, screen scraping techniques to automate workflows.	K3
4.	**Apply** Exception Handling techniques to ensure reliable bot execution.	K3
5.	**Understand** the process of deploying, managing, and updating RPA bots.	K2

SYLLABUS

UNIT-I (10Hrs)	**Introduction To Robotic Process Automation**: Scope and techniques of automation, Robotic process automation, Benefits of RPA, Components of RPA, RPA platforms, UiPath studio, UiPath Robot, UiPath orchestrator, the future of automation.
UNIT-II (8 Hrs)	**Record And Play**: UiPath Stack, Types of Robots, the user interface, Quick Access Toolbar, Different panels, Task recorder, Step by step examples using recorder. Sequence, Activities, Control flow, various types of loops, decision making, step by step example using sequence and flow chart, step by step example using sequence and control flow.
UNIT-III (10 Hrs)	**Data Manipulation and Controls**: Variables and Scope, Collections, Arguments, Data table usage, clipboard management, File operations, CSV/Excel to data table and vice versa. Finding and attaching windows, finding the control, techniques for a waiting for a control, act on controls, mouse and keyboard activities, working with UiExplorer, Handling events, Screen scraping, OCR.
UNIT-IV (10 Hrs)	**Assistant Bots, Exception Handling**: Assistant bots, Monitoring system event triggers, Hotkey trigger, Mouse trigger, System trigger Monitoring image and element triggers, Example of monitoring email, Example of monitoring a copying event and blocking it -

	Launching an assistant bot on a keyboard event. Exception Handling, common exceptions, logging and taking screenshots, Debugging techniques, collecting crash dumps, error reporting.
UNIT-V (8 Hrs)	**Deploying And Maintaining the Bot**: Publishing using publish utility, publish a workflow in Uipath, Overview of Orchestration Server, using orchestration server to control bots, using orchestration server to deploy bots, license management, publishing and managing updates. .

Textbooks:	
1.	Alok Mani Tripathi, "Learning Robotic Process Automation", Packt Publishing, 2018.
2.	Frank Casale, Rebecca Dilla, Heidi Jaynes, Lauren Livingston, "Introduction to Robotic Process Automation: a Primer", Institute of Robotic Process Automation,1st Edition 2015.
Reference Books:	
1.	Richard Murdoch, Robotic Process Automation: Guide to Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant", Independently Published, 1st Edition 2018.
2.	Srikanth Merianda," Robotic Process Automation Tools, Process Automation and their benefits: Understanding RPA and Intelligent Automation", Consulting Opportunity Holdings LLC, 1st Edition 2018.
3.	Lim Mei Ying, "Robotic Process Automation with Blue Prism Quick Start Guide: Create software robots and automate business processes", Packt Publishing, 1st Edition 2018.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AD4106	PE	3	--	--	3	30	70	3 Hrs.

REINFORCEMENT LEARNING

for AIDS

Course Objectives: This course is designed to:

1.	**To understand** the fundamental concepts and framework of Reinforcement Learning, including agents, environments, rewards, and policies.
2.	**To understand** the formulation and solution of the **multi-armed bandit problem** and action-value methods.
3.	**To understand** the structure and components of **Markov Decision Processes (MDPs)**.

Course Outcomes: Upon the completion of the course students will be able to:

S. No	Outcome	Knowledge Level
1.	**Understand and enumerate** the core elements of Reinforcement Learning, including agents, environments, rewards, policies, and their applications in real-world examples like Tic-Tac-Toe.	K2
2.	**Apply** action-value methods to solve multi-armed bandit problems, including nonstationary tracking, optimistic initialization, and contextual bandits.	K3
3.	**Apply** concepts of Finite Markov Decision Processes (MDPs) to compare value functions, returns, and optimal policies in episodic and continuing tasks.	K3
4.	**Understand** Monte Carlo methods for prediction, control, and off-policy learning, and discuss their role in solving real-world problems like scheduling and allocation.	K2
5.	**Understand** key applications and case studies of Reinforcement Learning, such as TD-Gammon, Acrobot, and job-shop scheduling.	K2

SYLLABUS

UNIT-I (10Hrs)	**The Reinforcement Learning Problem:** Reinforcement Learning, Examples, Elements of Reinforcement Learning, Limitations and Scope, An Extended Example: Tic-Tac-Toe, Summary, History of Reinforcement Learning.
UNIT-II (8 Hrs)	**Multi-arm Bandits:** An n-Armed Bandit Problem, Action-Value Methods, Incremental Implementation, tracking a Nonstationary Problem, Optimistic Initial Values, Upper-Confidence Bound Action Selection, Gradient Bandits, Associative Search (Contextual Bandits)
UNIT-III (10 Hrs)	**Finite Markov Decision Processes:** The Agent–Environment Interface, Goals and Rewards, Returns, Unified Notation for Episodic and Continuing Tasks, The Markov Property, Markov Decision Processes, Value Functions, Optimal Value Functions, Optimality and Approximation.

UNIT-IV (10 Hrs)	**Monte Carlo Methods:** Monte Carlo Prediction, Monte Carlo Estimation of Action Values, Monte Carlo Control, Monte Carlo Control without Exploring Starts, Off- policy Prediction via Importance Sampling, Incremental Implementation, Off- Policy Monte Carlo Control, Importance Sampling on Truncated Returns
UNIT-V (8 Hrs)	**Applications and Case Studies:** TD-Gammon, Samuel's Checkers Player, The Acrobot, Elevator Dispatching, Dynamic Channel Allocation, Job-Shop Scheduling.

Textbooks:	
1.	Richard S. Sutton and Andrew G. Barto, "Reinforcement Learning-An Introduction",2nd Edition, The MIT Press, 2018
2.	Marco Wiering, Martijn van Otterlo Reinforcement Learning: State-of-the Art (Adaptation, Learning, and Optimization (12)) 2012th Edition
Reference Books:	
1.	Vincent François-Lavet, Peter Henderson, Riashat Islam, An Introduction to Deep Reinforcement Learning (Foundations and Trends(r) in Machine Learning) , 2019
2.	Algorithms for Reinforcement Learning Author: Csaba Szepesvári
3.	Reinforcement Learning: A Survey (for theory clarity) Authors: Kaelbling, Littman, Moore

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM4106	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

HIGH PERFORMANCE COMPUTING

Common to AIDS & AIML

Course Objectives: This course is designed to:

1.	Understand parallel computing concepts, architectures, and performance issues.
2.	Design and analyze parallel algorithms for real-world problems.
3.	Implement parallel programs using modern platforms like OpenMP and CUDA.

Course Outcomes: Upon the completion of the course students will be able to:

S. No	Outcome	Knowledge Level
1.	**Explain** parallel programming platforms, implicit parallelism, and trends in microprocessor and architectural design.	K2
2.	**Apply** decomposition and mapping strategies to develop simple parallel algorithm models.	K3
3.	**Utilize** various communication patterns and basics of shared address space programming.	K3
4.	**List** performance metrics and analytical models to evaluate parallel programs.	K2
5.	**Apply** parallel algorithms for sorting and graph problems.	K3

SYLLABUS

UNIT-I (10Hrs)	**Introduction:** Motivating Parallelism, Scope of Parallel Computing, Parallel Programming Platforms: Implicit Parallelism, Trends in Microprocessor and Architectures, Limitations of Memory, System Performance, Dichotomy of Parallel Computing Platforms, Physical Organization of Parallel Platforms, Communication Costs in Parallel Machines, Scalable design principles, Architectures: N-wide superscalar architectures, multi-core architecture.
UNIT-II (8 Hrs)	**Parallel Programming:** Principles of Parallel Algorithm Design: Preliminaries, Decomposition Techniques, Characteristics of Tasks and Interactions, Mapping Techniques for Load Balancing, Methods for Containing Interaction Overheads, Parallel Algorithm Models, The Age of Parallel Processing, the Rise of GPU Computing, A Brief History of GPUs, Early GPU.
UNIT-III (10 Hrs)	**Basic Communication:** Operations- One-to-All Broadcast and All-to-One Reduction, All-to-All Broadcast and Reduction, All-Reduce and Prefix-Sum Operations, Scatter and Gather, All-to-All Personalized Communication, Circular Shift, Improving the Speed of Some Communication Operations. Programming shared address space platforms: threads-

	basics, synchronization, OpenMP programming.
UNIT-IV (10 Hrs)	**Analytical Models**: Sources of overhead in Parallel Programs, Performance Metrics for Parallel Systems, and the effect of Granularity on Performance, Scalability of Parallel Systems, Minimum execution time and minimum cost, optimal execution time. **Dense Matrix Algorithms**: MatrixVector Multiplication, Matrix-Matrix Multiplication.
UNIT-V (8 Hrs)	**Parallel Algorithms- Sorting and Graph:** Issues in Sorting on Parallel Computers, Bubble Sort and its Variants, Parallelizing Quick sort, All-Pairs Shortest Paths, Algorithm for sparse graph, Parallel Depth-First Search, Parallel BestFirst Search. **CUDA Architecture:** CUDA Architecture, Using the CUDA Architecture, Applications of CUDA Introduction to CUDA C-Write and launch CUDA C kernels, Manage GPU memory, Manage communication and synchronization, Parallel programming in CUDA- C.
Textbooks:	
1.	Ananth Grama, Anshul Gupta, George Karypis, and Vipin Kumar, "Introduction to Parallel Computing", 2nd edition, Addison-Wesley, 2003.
2.	Jason sanders, Edward Kandrot, "CUDA by Example", Addison-Wesley.
Reference Books:	
1.	Kai Hwang, "Scalable Parallel Computing", McGraw Hill 1998
2.	Shane Cook, "CUDA Programming: A Developer's Guide to Parallel Computing with GPUs", Morgan Kaufmann Publishers Inc. San Francisco,
3.	David Culler Jaswinder Pal Singh," Parallel Computer Architecture: A Hardware/ Software Approach", Morgan Kaufmann

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AD4108	SEC	--	1	2	2	30	70	3 Hrs.

PROMPT ENGINEERING

for AIDS

Course Objectives: This course is designed to:

1	Apply iterative prompting for clarity and context.
2	Create varied prompts to steer model outputs.
3	Construct chain-of-thought and structured prompts.
4	Develop retrieval-augmented pipelines to ground outputs.
5	Evaluate LLM agents and multimodal apps for ethics and robustness.

Course Outcomes: Upon the completion of the course students will be able to:

	Outcome	Knowledge Level
1.	**Explain** core principles of prompt engineering, its distinction from fine-tuning, and the iterative prompting lifecycle.	K2
2.	**Apply** advanced prompt patterns, such as few-shot, role-based, negative prompting, and constraint specification, to generate consistent and targeted LLM outputs.	K3
3.	**Construct** structured prompts for formatted outputs (e.g., JSON, tables) and chain-of-thought reasoning to solve complex tasks through decomposition.	K3
4.	**Develop** basic retrieval-augmented generation (RAG) pipelines using LangChain, including indexing, embeddings, and LCEL chains, to ground LLM responses in external data.	K2
5.	**Evaluate** LLM agents, multimodal applications, and prompts for ethics, robustness, security risks (e.g., prompt injection), and performance using manual and automated techniques.	K3

SYLLABUS

UNIT-I (10Hrs)	**Foundations of Prompt Engineering:** Definition of prompt engineering, Distinction between prompt engineering and model fine-tuning, Motivation and benefits of prompt engineering, Core principles of effective prompt design, Anatomy of a prompt, Setting up the Python environment for LLM interaction, Iterative prompting lifecycle, Common prompt pitfalls and remediation. **Lab Experiments:** Environment & Connectivity: Install required packages (e.g., transformers, openai); securely configure the API key; run a simple “Hello, world” prompt to verify model access. Baseline vs. Enhanced Prompts: Execute a naïve prompt (“Write a one-paragraph bio

	of Ada Lovelace.") and an enhanced prompt that adds role framing, specificity, and explicit format instructions; compare both outputs for relevance, completeness, and style. Iterative Refinement on a Simple Task: Summarize the plot of the Shakespearean play Romeo and Juliet in two sentences through three rounds of prompt tweaking: Minimal instruction. Addition of length and style constraints Specification of key content elements (setting and theme) Document how each iteration changes and improves the result. Diagnosing Prompt Failures & Edge Cases: Craft a vague or contradictory prompt; analyze the failure mode (ambiguity, missing context, or format errors); refine the prompt by adding examples or clarifying instructions.
UNIT-II (8 Hrs)	**Advanced Prompt Patterns & Techniques:** Enhanced prompt anatomy: contextual detail and explicit output specifications, Few-shot in-context prompting, Prompt structuring and template design, Role-based prompting to establish personas or system behavior, Negative prompting to filter or suppress undesired content, Constraint specification and instruction enforcement (e.g., length, format), Iterative prompt refinement and optimization. **Lab Experiments:** Few-Shot vs. Zero-Shot Comparison: Design and execute a zero-shot prompt and a few-shot prompt (with 2–3 exemplar input-output pairs) for a chosen text task (e.g., sentiment classification or translation); compare outputs for accuracy, consistency, and adherence to examples. Role-Based & Negative Prompting: Craft a role-based prompt to establish a specific persona (e.g., "You are a financial advisor…"); then create a negative prompt to suppress undesired content (e.g., "Do not mention any brand names"); evaluate how each influences the model's response. Constraint Specification & Iterative Refinement: Select an open-ended task (e.g., summarizing a technical article); issue a basic prompt; identify failures in length or format; refine the prompt by adding explicit constraints (word count, bullet format, etc.); document improvements over two refinement cycles.
UNIT-III (10 Hrs)	**Structured Output & Reasoning Techniques:** Importance of structured outputs for real-world applications, Promptingprompting for specific formats (lists, tables, Markdown), Generating valid JSON and YAML via explicit instructions, Eliciting chain-of-thought reasoning in zero-shot prompts, Decomposing complex tasks into manageable sub-tasks. **Lab Experiments:** Structured Format Prompting: Instruct the model to output information as bullet lists and Markdown tables (e.g., "List three benefits of daily exercise in a Markdown table with columns 'Benefit' and 'Description.'"); verify the output matches the requested structure.

	JSON/YAML Generation: Provide a brief dataset description (e.g., three books with title, author, publication year) and prompt the model to produce valid JSON or YAML; use a parser to validate syntax and refine the prompt if errors occur. Chain-of-Thought & Task Decomposition: Present a multi-step problem (e.g., a logic puzzle) and apply zero-shot CoT prompting (e.g., "Let's think step by step. Explain your reasoning before the final answer."); separately, decompose the problem into sequential sub-questions, collect partial answers, combine them, and compare accuracy against a direct-answer baseline. Retrieval-Augmented Generation (RAG), Overview of RAG architecture (indexing vs. retrieval + generation), Getting started with LangChain for LLM applications, Basics of LangChain Expression Language (LCEL), Simplified indexing pipeline: document loading & text splitting, Fundamentals of embeddings and vector stores, Building a basic retrieval-generation pipeline with an LCEL chain. **Lab Experiments:** Building a Simple LCEL Chain: Create a minimal LCEL script that accepts a fixed instruction (e.g., "Summarize this text: …"), passes it to an LLM, and prints the result; verify end-to-end execution. Basic Data Indexing for RAG: Load a small collection of documents; split into uniform chunks (e.g., 200 tokens); generate embeddings for each chunk; store them in an in-memory vector store; inspect for consistency. Constructing & Running a Basic RAG Chain: Build a pipeline that: Receives a user query Retrieves the top-k relevant chunks Constructs a combined prompt with context + query Send it to the LLM Returns the answer Test with sample queries and compare factual accuracy against a prompt without retrieval.
UNIT-V (8 Hrs)	**Agents, Multimodal AI & Ethical Evaluation:** Introduction to LLM agents and their basic architecture, Overview of multimodal AI models (VLMs), Prompting for text-to-image generation and image understanding, Importance of prompt evaluation beyond subjective judgment, Manual evaluation techniques (heuristic checks for accuracy, relevance, format), Introduction to "LLM-as-Judge" for automated evaluation, Security considerations (prompt injection, sensitive- information risks), Prompt-based mitigation strategies for safety and robustness, Ethical concerns (bias, misinformation, data privacy), Brief exploration of UI frameworks (Streamlit/Gradio) for deploying prompt-driven apps, Adapting to the evolving nature of prompt engineering through continuous learning. **Lab Experiments:** Building a Simple LLM Agent: Register a tool (e.g., a calculator function) and craft prompts that instruct the agent to invoke it when required; implement using LangChain or a function-calling API; test on queries requiring tool execution. Multimodal Prompting Exploration: Generate images from detailed text prompts; feed one generated image into an image-understanding model or API with an appropriate

	prompt; compare the returned caption to the original prompt to evaluate alignment. Prompt Evaluation & Ethics Workshop: Select two existing prompts and generate multiple outputs; apply manual heuristic checks for accuracy, relevance, and format compliance. Use an "LLM-as-Judge" prompt (e.g., "Rate these outputs on a scale of 1–5 for clarity and correctness.") to automate evaluation. Design a prompt- injection test (e.g., "Ignore previous instructions…"), observe the response, then refine system prompts to mitigate the vulnerability.

Textbooks:	
1.	James Phoenix, Mike Taylor, "Prompt Engineering for Generative AI", O'Reilly, To Release in May 2024 https://www.oreilly.com/library/view/prompt-engineering-for/9781098153427/
2.	Gilbert Mizrahi, "Unlocking the Secrets of Prompt Engineering: Master the Art of Creative Language Generation to Accelerate Your Journey from Novice to Pro", January 2024 https://www.packtpub.com/en-in/product/unlocking-the-secrets-of-prompt-engineering9781835083833
Online Learning Resources:	
1.	Michael Ferguson, "Prompt Engineering: The Future of Language Generation", January 2023 https://books.apple.com/us/book/prompt-engineering-the-future-of-languagegeneration/id6445529200
2.	"Prompt Engineering Guide", https://www.promptingguide.ai/
3.	"Prompt Engineering for Generative AI", Google, https://developers.google.com/machinelearning/resources/prompt-eng

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC4101	**MC**	**2**	**--**	**--**	**--**	**30**	**-**	**3 Hrs.**

CONSTITUTION OF INDIA

(Common to all)

Course Objectives: This course is designed to:

1	To provide understanding of the fundamental principles, sources, and key features of the Indian Constitution.
2	To familiarize students with the structure and functioning of Union, State, and Local Governments in India.
3	To develop awareness about the roles of constitutional bodies like the Election Commission and welfare commissions in promoting democracy and social justice.

Course Outcomes: Upon the completion of the course students will be able to:

S.No	Outcome	Knowledge Level
1.	**Apply** the concepts of the Indian Constitution to interpret its features such as Citizenship, Preamble, Fundamental Rights and Duties, and Directive Principles of State Policy.	**K3**
2.	**Apply** the structure and functions of the Union Government to analyze the roles of executive, legislature, and judiciary in the Indian administrative system.	**K3**
3.	**Apply** the structure and functions of the State Government to interpret the roles of the Governor, Chief Minister, and State Secretariat in administration.	**K3**
4.	**Apply** the structure and functions of local administration to interpret the roles of district, municipal, and Panchayati Raj institutions in grass roots governance.	**K3**
5.	**Apply** the roles and functions of the Election Commission and various welfare commissions to interpret their significance in ensuring democratic governance and social justice.	**K3**

SYLLABUS

UNIT-I (10Hrs)	Introduction to Indian Constitution: Constitution meaning of the term, Indian Constitution - Sources and constitutional history, Features - Citizenship, Preamble, Fundamental Rights and Duties, Directive Principles of State Policy.
UNIT-II (8 Hrs)	Union Government and its Administration Structure of the Indian Union: Federalism, Centre-State relationship, President: Role, power and position, PM and Council of ministers, Cabinet and Central Secretariat, Lok Sabha, Rajya Sabha, The Supreme Court and High Court: Powers and Functions.
UNIT-III (10 Hrs)	State Government and its Administration Governor - Role and Position - CM and Council of ministers, State Secretariat: Organisation, Structure and Functions.

UNIT-IV (10 Hrs)	Local Administration - District's Administration Head - Role and Importance, Municipalities — Mayor and role of Elected Representative - CEO of Municipal Corporation Pachayati Raj: Functions PRI: Zilla Panchayat, Elected officials and their roles, CEO Zilla Panchayat: Block level Organizational Hierarchy - (Different departments), Village level - Role of Elected and Appointed officials - Importance of
	grass root democracy.
UNIT-V (8 Hrs)	Election Commission: Election Commission- Role of Chief Election Commissioner and Election Commissionerate State Election Commission: Functions of Commissions for the welfare of SC/ST/OBC and women.

Textbooks:	
1.	Durga Das Basu, Introduction to the Constitution of India, Prentice — Hall of India Pvt. Ltd..New Delhi.
2.	Subash Kashyap, Indian Constitution, National Book Trust.
Reference Books:	
1.	J.A. Siwach, Dynamics of Indian Government & Politics, Sterling Publishers.
2.	D.C. Gupta, Indian Government and Politics, S Chand.
3.	H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes, Universal Law Publication.
4.	M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice Hall of India Pvt. Ltd.. New Delhi.
5.	Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press 2012.
e-Resources	
1.	https://onlinecourses.swayam2.ac.in/e-learning/preview/imb26_mg151
2.	https://onlinecourses.nptel.ac.in/noc24_lw05/preview
3.	https://nptel.ac.in/courses/129106750

Regulation:R23	IV/IV-B.Tech. II -Semester								
ARTIFICIAL INTELLIGENCE & DATA SCIENCE									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	**Course Name**	**Category**	**L**	**T**	**P**	**Cr**	**C.I.E.**	**S.E.E.**	**Total Marks**
B23AD4201	Project and Internship	**PR**	**-**	**-**	**24**	**12**	**60**	**140**	**200**

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AD4201	**PR**	**--**	**--**	**24**	**12**	**60**	**140**	**3 Hrs.**

PROJECT WORK

(For AIDS)

Course Objectives: This course is designed to:

1	To enable students to independently research, analyze, and formulate a well-defined engineering problem by reviewing contemporary literature and technical databases.
2	To guide students in conceptualizing, designing, and optimizing innovative solutions, components, or processes while balancing realistic socio-economic and technical constraints.
3	To train students to build, test, and troubleshoot physical prototypes or software systems through the self-taught application of modern engineering tools and simulation software.
4.	To encourage students to evaluate the environmental, safety, legal, and cultural impacts of their engineering solutions while maintaining professional ethics.
5.	To nurture collaborative teamwork, project resource management, technical report writing, and effective multi-media presentation skills.

Course Outcomes: Upon the completion of the course students will be able to:

	Outcome	Knowledge Level
1.	**Identify and formulate** a complex engineering problem by reviewing literature and conducting feasibility studies.	**K4**
2.	**Design and develop** an innovative, safe, and sustainable solution, component, or process using modern engineering tools.	**K5**
3.	**Implement and test** the designed solution, analyzing the resulting data to draw valid engineering conclusions.	**K4**
4.	**Evaluate** the impact of the project solution on society, environment, and health, incorporating ethical considerations.	**K5**
5.	**Communicate** effectively through technical reports, presentations, and project demonstrations.	**K3**
6	**Demonstrate** teamwork, leadership, and project management skills, adhering to ethical practices and financial constraints.	**K3**

*The object of Project Work is to enable the student to take up investigative study in the broad field of, **Artificial Intelligence & Data Science** either fully theoretical/practical or involving both theoretical and practical work to be assigned by the Department on an individual basis or a group of students, under the guidance of a Supervisor. This is expected to provide a good initiation for the student(s) in R&D work.

The assignment to normally include:

Survey and study of published literature on the assigned topic.

Working out a preliminary approach to the problem relating to the assigned topic. Conducting preliminary Analysis/Modeling/Simulation/Experiment/Design/ Feasibility. Preparing a written report

on the study conducted for presentation to the department. Final Seminar, as oral Presentation before a departmental committee.